M.Tech Project Dissertation

on

Development of Digital Twin for Warehousing

Master of Technology

Stage - I Project Report

by

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Declaration

I affirm that this written submission is a representation of my own ideas, expressed in my own words. Wherever I have used the ideas or words of others, I have appropriately cited and referenced the original sources. I confirm that I have accurately acknowledged all sources used in the creation of this report. I also affirm that I have upheld all standards of academic integrity and honesty, ensuring that no idea, data, fact, or source has been misrepresented, fabricated, or falsified in this submission. I am aware that any breach of these principles may result in disciplinary action by the Institute and could also lead to legal consequences if sources are not properly cited or permissions were not obtained when required.

Name: Saimadhav S



Signature:

Date: October 30, 2026

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*Dedicated to my parents and teachers (*gurus*)*

Abstract

Vision systems deployed in aerial, agricultural, and outdoor environments often degrade under adverse atmospheric conditions such as fog, smog, blizzards, or sandstorms. These effects cause significant distribution shifts in image statistics, reducing the reliability of vision transformers and multimodal encoders trained on clear-weather datasets.

This work explores diffusion-based generative frameworks to synthetically recreate such environmental conditions, enabling vision models to adapt and generalize effectively to real-world scenes.

A focused dataset was curated containing images of extreme atmospheric domains such as fog, smog, sandstorm, forest fire, storm, and underwater to evaluate existing, adaptive, and proposed solutions under consistent conditions.

Foundational diffusion architectures, including U–Net, DDPM and Conditional Diffusion were implemented to understand their generative dynamics. Building on these foundations, style-adaptive frameworks such as StyleShot, CSGO and InstantStyle were evaluated on the curated dataset using state-of-the-art vision transformers to assess their ability to reproduce physically consistent style shifts.

An existing model, **InstantStyle-Plus**, was adapted by extending InstantStyle with ResNoise and IP-Adapter modules for semantic and structural consistency, and further enhanced through a **Contrastive Style Descriptor (CSD)** guidance loop for perceptually aligned, distribution-shift–aware image synthesis .

Finally, a novel diffusion system integrating **InstantStyle with CSD loss** was introduced, guiding the denoising trajectory through contrastive style supervision to achieve physically realistic, depth-aware stylization while preserving semantic fidelity. The proposed framework demonstrates superior perceptual realism and environmental consistency, producing high-quality, style-shifted datasets that enhance the robustness of vision models for agricultural, aerial, and environmental monitoring applications.

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Chapter 1

Introduction

Operations of a warehouse are currently monitored and taken actions by a warehousing professional. With Digital Twin, there is scope for automation. The decision-making at various levels can be automated. Digital Twin for a warehouse paves the way forward to automate decisions at various levels including Strategic, Tactical and Operational levels.

1.1 Problem Statement

State of the Art in Digital Twin for Warehouse Operations Literature, stop at the Digital Model or Digital Shadow level. This work aims to develop a fully integrated Digital Twin for Warehouse Operations. The idea is to first characterise a model for a warehouse scenario, followed by implementing an Opti-State Control approach to stay in the Optimal State. The Deadheading scenario in a warehouse is taken as the reference scenario. Then the goal is to continually remain in the optimal state, given the disturbances encountered by the warehouse.

Chapter 2

Literature Survey

This chapter builds the theoretical foundation for the work presented in this report.

2.1 Warehouse Digital Twins

The first paper reviewed, authored by Ferrari et al. [**?**], has focused on the creation of a Digital Shadow for an Automated Storage and Retrieval System (AS/RS) in a university logistics lab. A detailed simulation model was developed using AnyLogic, incorporating discrete event and agent-based modeling. Although real-time control was not achieved, the model was able to provide insights into system layout and performance. A Warehouse Management System (WMS) and a Warehouse Control System (WCS) were used to coordinate operations such as order generation and storage location management. The system was designed to simulate the behavior of physical operations closely, though without live data feedback into the physical system.

The approach by Lugaresi et al. [**?**], involved a small-scale manufacturing setup using LEGO components. A closed-loop production system with four stations (S1, S2, S3, S4) and associated buffers was designed, controlled by LEGO® MINDSTORMS® EV3® bricks programmed with Python scripts. Two product types (A and B) with distinct processing times, modeled using Weibull distributions, were processed, and setup times for switching between product types were required by stations S2, S3, and S4, with S2's setup time increasing exponentially after three consecutive type A pallets. A discrete event simulator, developed in ARENA® (Sim.doe), was used as the digital model, synchronized with the physical LMS through data exchange files (*.txt) containing buffer levels, processing times, and production status. Real-time data from the LMS were collected, the digital model was updated, and alternative production policies were simulated to identify the optimal response to a disruption in the RTS framework.

In the work by Zhang et al, [**?**] the core idea is *opti-state control*. Rather than a single optimization before execution, the system is continuously monitored. As the warehouse is deviated from its desired performance, there is adaptive reconfiguration

of plans. For warehouse and intralogistics research, the paper's contribution is to move from digital twins as a monitoring tool, toward a closed-loop control system.

Summary

Research has steadily advanced toward controllable and interpretable generation. Transformers introduced attention mechanisms, ViTs extended them to vision, and GANs and VAEs pioneered generative modeling. Diffusion unified these ideas into a stable probabilistic framework, enhanced by latent conditioning, cross-attention, and adapters like ControlNet and IP-Adapter. Contrastive systems such as CLIP and CSD added semantic and stylistic grounding, forming the basis for the adaptive, CSD-guided diffusion explored here.

Chapter 3

Foundational Experiments

This chapter presents some foundational experiments conducted to get used to the sensing technologies needed as part of the Digital Twin.

3.1 RFID Experiments

A contactless technology called RFID (Radio Frequency Identification) is utilized for data collection and automatic identification. It eliminates the need for physical contact by using radio waves to transport data between an RFID tag and a reader. RFID tags and RFID readers are the two fundamental parts of an RFID system. A tiny electronic device called an RFID tag is affixed to the item you wish to track or identify. The tag is made up of an antenna that connects to the reader and a microchip that saves data. Active and passive RFID tags are two different categories.

Based on the MFRC522 chip created by NXP Semiconductors, the RC522 RFID module is a popular and inexpensive RFID reader/writer module. The standard frequency for RFID communication, 13.56 MHz, is where it operates. This module supports several communication protocols, such as SPI (Serial Peripheral Interface), I2C (Inter-Integrated Circuit), and UART (Universal Asynchronous Receiver-Transmitter), and is appropriate for short-distance communication, usually within a 5 cm range. The majority of RC522 modules include:

- A 1K RFID card from MIFARE Classic
- A key fob or tag

3.2 PIR sensor experiments

An electronic sensor that detects infrared (IR) light emitted by objects within its field of vision is called a passive infrared sensor (PIR sensor). PIR-based motion detectors are the most common application for them. PIR sensors are frequently utilized in automated lighting and security alarm systems.

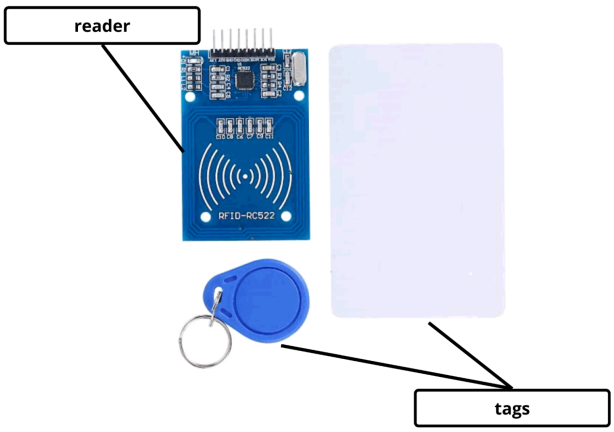


Figure 3.1: RFID tag and reader

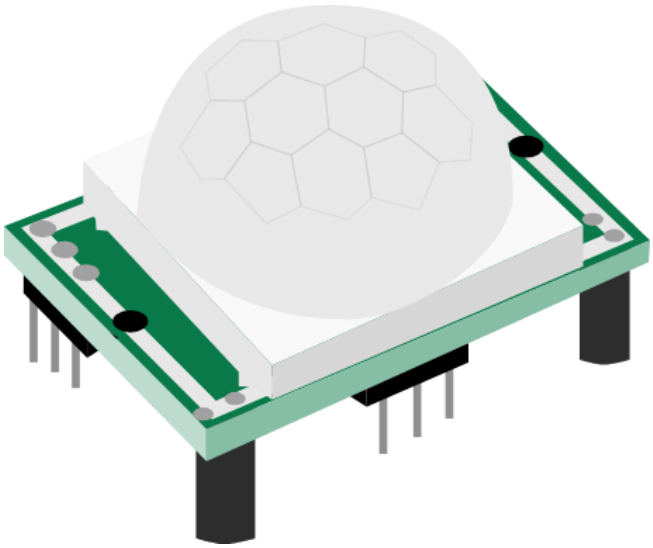


Figure 3.2: PIR sensor

3.3 Ultrasonic Sensor Experiments

The ultrasonic sensor uses sound waves and the echo principle to calculate distance from obstacles. Distance Travelled = $\frac{Speed of Sound \times Time Travelled}{2}$ HCSR04 is the code for an ultrasonic sensor module avaialable in the market. It has a range from 2cm to 400cm. The ranging accuracy is 3mm. The sensor module has four pins: VCC, GND, TRIG and ECHO.

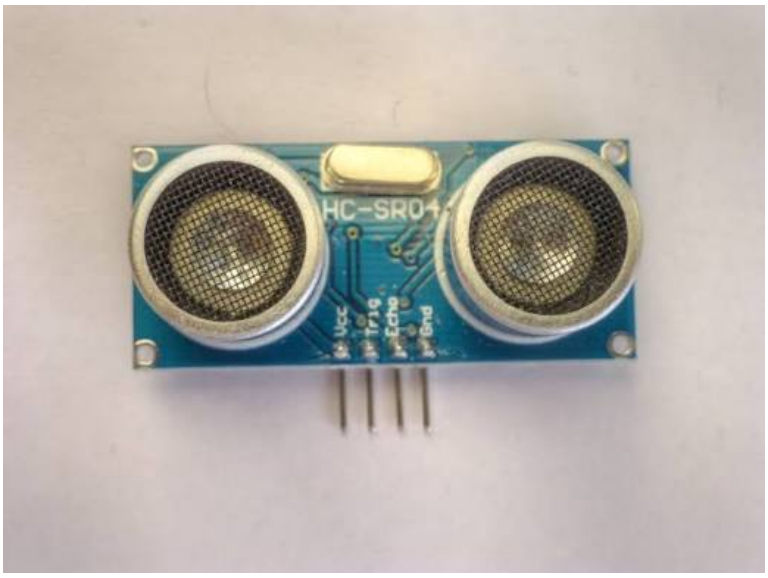


Figure 3.3: Ultrasonic sensor module

3.4 Summary of Foundational Experiments

Through the above experiments, an idea of different Physical to Virtual (P2V) experiments was evaluated.

Chapter 4

Characterisation of a model

4.1 Deadheading problem and the Single Cycle protocol

The material handling device proceeding or returning empty while attending to an order is the deadheading problem. By default or following the single cycle protocol, when travelling to pick the item, the forklift is empty. Also, when travelling back after storing an item, the forklift is empty.



Figure 4.1: Single Cycle Protocol

4.2 Dual Cycle Protocol

Enter, the dual cycle protocol. In the dual cycle protocol, a picking operation is paired with a putting (stowing) operation. This way while returning after the stow operation, the picker proceeds to the pick operation, thereby never going empty handed.

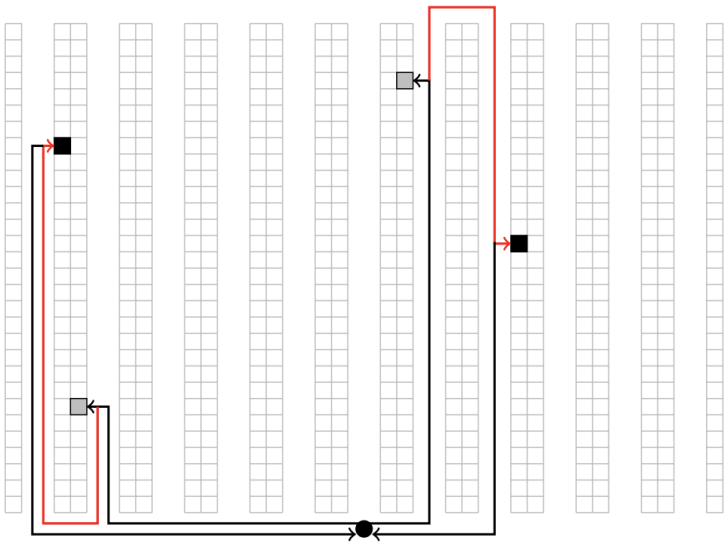


Figure 4.2: Dual Cycle Protocol

The Dual cycle protocol relies on the following optimization model:

- M is set of locations covered by store operations.
- N is set of locations covered by picking operations.
- $i \in M$ and $j \in N$
- d_{ij}: distance between locations i and j
- x_{ij}: 1 if store operation at i is succeeded by pick operation from j, 0 otherwise

$$
\begin{aligned}
\min \quad & \sum_{i,j} d_{ij} x_{ij} \\
\text{s.t.} \quad & \sum_{i} x_{ij} = 1 \qquad \forall j \\
& \sum_{j} x_{ij} = 1 \qquad \forall i \\
& x_{ij} \in \{0,1\} \qquad \forall i,j
\end{aligned}
$$

Summary

Chapter 5

Comparative Analysis

This chapter summarizes the comparative results across all evaluated models — StyleShot [**?**], CSGO [**?**], InstantStyle [**?**], the adapted InstantStyle-Plus [**?**], and the proposed InstantStyle [**?**] + CSD [**?**] Guidance — using both qualitative and quantitative indicators. The goal was not to maximize classification accuracy but to assess the **extent and realism of distribution-shift generation** under adverse environmental styles.

5.1 Qualitative Comparison

Representative outputs from the curated dataset are shown in Figure 5.1. Across five reference prompts (*tractor plowing field*, *cornfield horizon*, *sheep herd*, *greenhouse*, and *flooded crop area*), the progressive improvement in visual realism and depth consistency is clearly visible.

Observations:

- **StyleShot** [**?**]**:** Produced overly artistic and painterly generations. While visually appealing, it performed poorly at style transfer, merely recoloring the scene without volumetric haze or realistic scattering. The model lacks depth-aware modulation, making its outputs aesthetically strong but physically implausible.
- **CSGO** [**?**]**:** Improved overall tonal blending but still lacked volumetric realism. Edges remained unnaturally sharp, and atmospheric style appeared as global filters rather than media that wrap around objects.
- **InstantStyle** [**?**]**:** Delivered the most balanced realism among existing systems, achieving physically faithful haze and turbidity while preserving prompt semantics and stable color consistency.
- **InstantStyle-Plus** [**?**]**:** Enhanced spatial and semantic stability through ControlNet [**?**] integration, creating smoother volumetric cues and reducing chromatic drift, though sometimes slightly softening contrast.

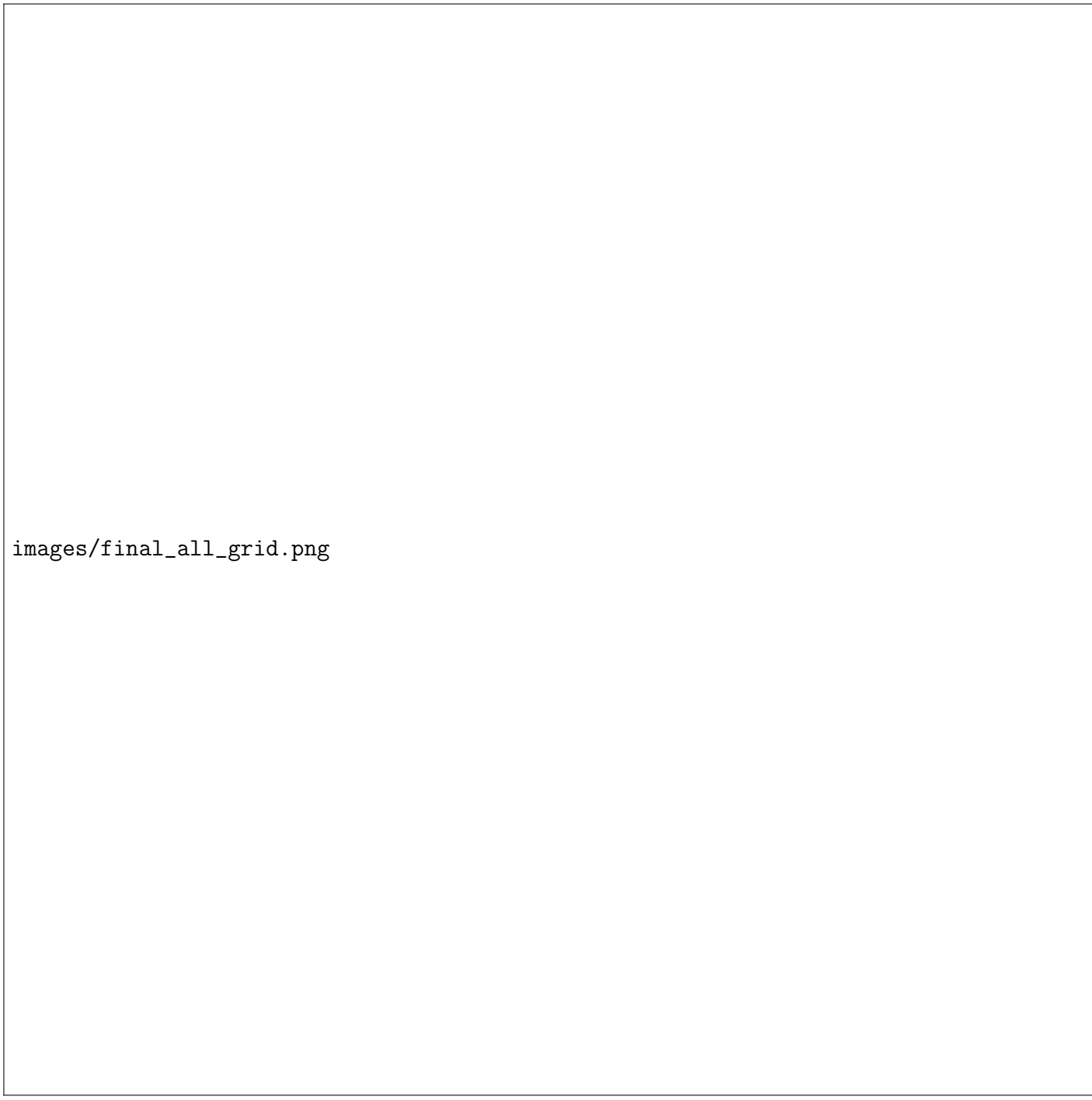


Figure 5.1: Qualitative comparison (five representative samples) across StyleShot, CSGO, InstantStyle, InstantStyle-Plus, and Proposed model. The proposed method achieves realistic environmental veiling and depth-consistent haze while preserving semantic integrity.

- **Proposed (InstantStyle [?] + CSD [?] Guidance):** Combined all strengths—strong volumetric realism, perceptually coherent tone alignment, and stable structure—with minimal added computation.

5.2 Accuracy Behavior Under Distribution Shift

To measure the perceptual domain shift caused by style generation, a Vision Transformer [**?**] classifier (ViT-B/16 trained on clear-weather images) was evaluated on the stylized outputs. A lower classification accuracy implies that the generated images are sufficiently distinct from the source domain—indicating a stronger, more realistic distribution shift.

Table 5.1: Classification accuracy (%) of a ViT-B/16 model on stylized images. Lower accuracy implies stronger and more realistic domain shift, provided visual quality remains high.

Model	**ViT Accuracy (↓)**
InstantStyle	62.7
InstantStyle-Plus	50.9
StyleShot	49.0
CSGO	47.8
Proposed (InstantStyle + CSD)	**30.3**

The results align with qualitative observations. **InstantStyle** achieves the best balance of realism and prompt consistency, while **InstantStyle-Plus** and the **Proposed CSD-guided** method induce a significantly stronger and more physically meaningful domain shift, reducing ViT accuracy by nearly half. In contrast, **StyleShot** achieves a similar drop in accuracy but due to over-artistic distortion, not realistic environmental transfer—its outputs deviate semantically and physically from real-world phenomena. Thus, the **Proposed (InstantStyle + CSD)** model produces the most convincing style shift while maintaining photorealism.

5.3 Parameter Efficiency

A model's practicality depends not only on realism but also on computational footprint. Table 5.2 summarizes parameter statistics for each framework
Despite integrating perceptual feedback, the proposed method retains nearly the same parameter count as InstantStyle, confirming that **quality improvements stem from perceptual correction, not additional learnable modules**. This efficiency makes it particularly suitable for real-time or low-resource generative applications.

Table 5.2: Parameter comparison across evaluated frameworks.

Model	Total Params (Billion)	Trainable (Million)
StyleShot	2.1–2.2	435–585
CSGO	6.6–6.7	650–760
InstantStyle	6.0	22
InstantStyle-Plus	7.2–7.5	600–700
Proposed (CSD)	6.6–6.7	**22**

5.4 Insights

From both qualitative and quantitative perspectives, the models reveal clear trade-offs:

- **StyleShot [?]:** Generates artistic, stylized outputs but performs poorly in physical style realism. The style overlay lacks volumetric cues, producing attractive but unrealistic results.
- **CSGO [?]:** Provides explicit modularity and stable fusion but requires content-based supervision to achieve realism; without it, results remain shallow.
- **InstantStyle [?]:** Demonstrates physically accurate scattering and atmospheric realism with excellent prompt alignment—strong generation and moderate style shift.
- **InstantStyle-Plus [?]:** Enhances structure preservation and environmental stability through ControlNet [?] conditioning, producing realistic depth-aware cues.
- **Proposed (InstantStyle [?] + CSD [?] Guidance):** Introduces perceptual supervision that strengthens volumetric realism and style coherence while keeping parameters low, achieving the most balanced performance overall.

Overall, while lower classification accuracy corresponds to a stronger domain shift, true success lies in combining perceptual quality and physical realism. The proposed method satisfies both — generating style-shifted scenes that visually resemble natural atmospheric variations while remaining semantically faithful to the input prompt.

5.5 Summary

The comparative analysis confirms that the **InstantStyle [?] + CSD [?] Guidance** framework achieves the most realistic and perceptually stable environmental stylization. It generates distribution-shifted domains that challenge vision classifiers without sacrificing photorealism or structural integrity, outperforming all baselines in both realism and efficiency.

Chapter 6

Conclusion and Future Work

6.1 Conclusion

This study explored the evolution of diffusion-based style transfer frameworks, beginning from foundational experiments on architectures like Vision Transformers [**?**], U–Nets [**?**], and DDPMs [**?**], and gradually advancing toward adaptive and perceptually guided diffusion systems. The focus was to understand how these models respond to distribution shifts caused by environmental styles such as fog, haze, snow, sandstorms, and underwater scattering. For comparitive Analysis extreme shifts were considered and through this exploration, it was observed that early frameworks like StyleShot [**?**] and CSGO [**?**], though capable of stylization, lacked volumetric realism and semantic consistency. InstantStyle [**?**] emerged as a more stable and physically coherent framework, offering controllable style injection through its IP-Adapter [**?**] mechanism. Building upon this, the adapted InstantStyle-Plus [**?**] added spatial consistency using ControlNet [**?**], while the proposed InstantStyle with CSD [**?**] Guidance integrated perceptual feedback directly into the diffusion process. This final model demonstrated the best balance between realism, controllability, and computational efficiency, producing physically consistent, depth-aware, and semantically aligned outputs. The results highlight that realistic domain adaptation through diffusion models is best achieved not by increasing complexity, but by combining adaptive structure with perceptual understanding.

6.2 Future Work

Future work can extend this research toward more physics-aware and adaptive generative pipelines. Integrating explicit depth or scattering priors into diffusion sampling could improve the realism of environmental media like fog and haze. Developing dynamic guidance schedules that adjust style strength automatically across scenes may further enhance generalization. Testing on large-scale classification and detection tasks will confirm whether these synthetic shifts enhance real-world robustness.

